AMD Power Gadget

User Manual & Comprehensive Feature Guide

Version 3.15.0

Introduction

Welcome to **AMD Power Gadget** and **SMCAMDProcessor**. This suite provides comprehensive telemetry and power management capabilities for AMD Ryzen processors on macOS (Hackintosh).

This manual explains *every single option, slider, and button* available in the application, leaving nothing to guesswork.

1. System Requirements & OpenCore Setup

1.1 Essential Kexts

Ensure the following kexts are present in your `EFI/OC/Kexts` folder and injected in your `config.plist` under `Kernel -> Add`, in this exact order:

1. `Lilu.kext` (Must be first)
2. `VirtualSMC.kext` (SMC emulator - DO NOT use FakeSMC)
3. `AMDRyzenCPUPowerManagement.kext` (Provides raw CPU data and SuperIO access)
4. `SMCAMDProcessor.kext` (Exports data to VirtualSMC for tools like iStat Menus)

1.2 OpenCore Quirks & Boot Arguments

- **ProvideCurrentCpuInfo** (Kernel → Quirks): Set to `True`. Mandatory for macOS to correctly map AMD core topologies.
- **agdpmod=pikera** (boot-args): Required for Radeon RX 6000 series (Navi) to prevent black screens.

1.3 SMCAMDProcessor Custom Boot Arguments

You can pass these optional boot arguments in OpenCore (`NVRAM -> boot-args`) to unlock advanced functionality or debug the kexts:

- `-amdpnopchk`: Disables privilege checks. Required if you want to use the "Directly Edit Raw P-State Registers" feature in the app without running it as root.
- `-amdcppcactive`: Forces CPPC (Collaborative Processor Performance Control) into Active Mode instead of Autonomous Mode, giving the macOS kernel more direct control over frequency selection rather than relying entirely on the SMU.
- `-amdpdbg`: Enables verbose debug logging for `AMDRyzenCPUPowerManagement.kext`. Useful for troubleshooting kernel panics or missing sensor data.

2. Power & Frequencies Tab

This tab provides real-time monitoring of your CPU's internal metrics.

2.1 Core Metrics & Silicon Quality

- **Core Frequencies**: Displays the real-time clock speed of each physical and logical core.
- **Silicon Quality Ranking (1. ~ X.)**: Zen 3/4 processors evaluate the quality of the silicon on a per-core basis. The application reads these CPPC tags and ranks your cores. Core `1.` is your absolute best core, capable of sustaining the highest boost frequencies at the lowest voltages. Use this data when tweaking the Curve Optimizer.
- **Package Power Tracking (PPT)**: Shows the total wattage consumed by the CPU package in real time.
- **Tctl / Tdie Temperatures**: The absolute junction temperature of the CPU.

3. Profiles Tab (CPU Speed Management)

The Profiles tab allows you to fundamentally alter how the CPU scales its frequencies and voltages.

3.1 EPP Profiles (Hardware Autonomous)

Energy Performance Preference (EPP) relies on the CPPC (Collaborative Processor Performance Control) interface. Instead of macOS dictating frequencies, you provide a "hint" to the CPU's internal SMU, and the hardware scales autonomously based on live load.

- **Power Saver**: Heavily limits boost clocks to prioritize battery life and acoustics.
- **Balanced**: The default state. Boosts when needed, but aggressively downclocks during idle.
- **Performance**: Keeps the cores highly responsive, maintaining higher base clocks and prioritizing single-thread boost speed over power efficiency.

3.2 CPU Speed Profiles (Legacy / Manual P-State)

Older Ryzen processors (or specific manual tuning scenarios) rely on static P-States (Power States).

- **Manual P-State Override**: Selecting a profile here completely restricts the CPU to a specific P-State step (e.g., P0 for max performance, P2 for base clock). The CPU will NOT scale dynamically; it is locked to the selected step.
- **Directly Edit Raw P-State Registers**: A highly advanced feature. Clicking this allows you to manually input the Hex values for Frequency, Voltage (VID), and DID.

  > **[WARNING]** Requires kext privilege checks disabled. Incorrect VID values will cause instant system shutdown or potential hardware degradation.

4. Advanced CPU Tuning: Curve Optimizer

The **Curve Optimizer** is the most powerful tool for AMD Ryzen users. It adjusts the voltage/frequency curve dynamically.

- **Per-Core Curve Offsets (-30 to +30)**: Instead of applying a blanket voltage offset, you can assign a unique offset to *each individual physical core*.
- **How it works**: A negative offset (e.g., `-15`) tells the CPU to use less voltage for a given frequency. Because the CPU runs cooler at a lower voltage, the Precision Boost algorithm automatically allows it to sustain higher boost clocks for longer periods.
- **Strategy**: Apply larger negative offsets (e.g., `-25`) to your lowest-ranked cores, and smaller offsets (e.g., `-10`) to your top-ranked cores (which are already pushed to their voltage limits from the factory).

5. Fan Control Tab

This tab interfaces with your motherboard's SuperIO controller (e.g., ITE IT8686E) to manage case and CPU fans.

5.1 Fan Monitoring & Hiding

- **Fan List**: Shows all detected fans, their RPM, and PWM Throttle percentage.
- **Hide Fan (Eye Slash Icon)**: If your motherboard reports ghost fans or headers like `H_AMP` show erratic values (e.g., `40 RPM` when disconnected), click the eye slash icon to permanently hide it from the UI and Menu Bar.
- **Show All (X hidden)**: Restores any hidden fans.

5.2 Quick Presets

- **All Auto (Arrow Circle Icon)**: Returns control of all fans back to the motherboard's BIOS logic.
- **Max Speed (Wind Icon)**: Instantly forces 100% PWM duty cycle on all connected fans (Take Off mode).

5.3 Closed-Loop Custom Fan Curves & Protection

Toggle **"Dynamic Next-Gen Fan Curves"** to override the BIOS and manage fans entirely via the macOS kernel.

- **Interactive Editor**: Drag the points on the graph to define your custom temperature-to-PWM curve.
- **256-Step LUT Interpolation**: The software translates your curve into a smooth 256-step Look-Up Table.
- **Smooth Ramping (Hysteresis)**: The kernel evaluates the curve with hysteresis to prevent fans from aggressively ramping up and down during sudden, momentary CPU temperature spikes.

5.4 GPU Fan Control (Zero RPM / SPPT)

As stated in the UI, direct software-based GPU fan speed overrides (like drawing curves for your Radeon RX 6800 XT) are **not supported** by the macOS kernel for AMD GPUs.

- **The Solution**: You must extract your vBIOS, use **MorePowerTool (MPT)** on Windows to modify the acoustic limits and Zero RPM toggles, and inject the resulting Soft PowerPlay Table (SPPT) string into OpenCore's `config.plist` under `DeviceProperties`. The UI provides direct links to the SPPT Guide and MPT downloads.

6. Menu Bar Extra & Preferences

The Menu Bar Extra provides a highly customizable, compact view of your system's vitals directly in the macOS menu bar.

6.1 Display Configuration

Click the AMD Power Gadget icon → **Preferences**:

- **Include CPU/GPU/Fans**: Toggle specific widgets on or off.
- **Peak Tracking**: The Menu Bar dropdown automatically logs the **Peak** and **Minimum** values for all your selected metrics during the current session.

6.2 Application Settings

- **Update Interval**: Adjust how frequently the sensors are polled. 1-second intervals provide real-time accuracy; higher intervals save CPU cycles.
- **Theme**: Force Dark Mode, Light Mode, or respect System settings (utilizes Liquid Glass vibrancy).

7. Driver Dump Utility (Advanced)

If your fans are reading incorrect RPMs, it usually means the 16-bit tachometer registers or the internal clock divisors are misaligned for your specific SuperIO chip revision. Run `Tools/IT8686E_Dump.sh` as `sudo` to output the raw Hex values of the Environment Controller (EC) registers. Use this to verify if the raw byte from the `FAN_RPM_REGS` needs a multiplier adjustment in the kext source code.